Use of Phenolic Resin Composites as Friction Materials with Zinc Oxide Whisker Reinforcement

1. Introduction to Phenolic Resin Friction Materials

Phenolic resins serve as the primary binder matrix in most organic friction materials due to their excellent heat resistance, dimensional stability, and ability to wet and bond various fillers and reinforcements . In friction applications such as brake pads, clutch facings, and synchronizer rings, phenolic resin composites must provide stable friction coefficients, adequate wear resistance, and thermal stability under repeated sliding contact. However, unmodified phenolic resins are inherently hard and brittle, which can lead to noise, chatter, and inadequate wear performance in demanding applications .

2. Zinc Oxide Whiskers as Functional Fillers

2.1 Unique Morphology and Properties

Tetrapod-shaped zinc oxide whiskers (T-ZnOw) represent a distinctive class of reinforcing fillers characterized by a three-dimensional, four-needle structure radiating from a central core . This geometry distinguishes them from conventional one-dimensional whiskers such as potassium titanate or aluminum silicate whiskers, which are linear in form.

The unique single-crystal structure of T-ZnOw provides several functional benefits when incorporated into phenolic resin matrices:

- **Isotropic reinforcement**: The three-dimensional shape distributes stress isotropically throughout the matrix, unlike fibrous reinforcements that provide strength primarily along their axis

- **Micro-reinforcement effect**: The high modulus and strength of the single-crystal legs resist external forces and improve wear resistance

- **Thermal conductivity**: T-ZnOw exhibits thermal conductivity of approximately 25.3 W/m·K, which helps dissipate frictional heat

- **Antistatic capability**: The three-dimensional network can eliminate static charge generated during friction, reducing ozone formation and material degradation

2.2 Mechanisms of Property Enhancement

When dispersed in phenolic resin, T-ZnOw acts through several mechanisms to improve friction material performance:

Mechanical Interlocking: The tetrapod legs create mechanical anchoring points within the cured resin matrix, improving cohesion and resistance to shear forces during braking .

Crack Deflection and Energy Dissipation: The three-dimensional structure deflects propagating cracks and dissipates energy through whisker pull-out and fracture, preventing catastrophic failure .

Thermal Management: The thermally conductive whiskers create heat dissipation pathways that reduce surface temperature during friction, inhibiting the formation of self-lubricating films that can destabilize friction coefficients .

Noise Reduction: By disrupting the homogeneity of the friction surface and promoting uniform contact, T-ZnOw can eliminate friction-induced vibration and noise .

3. Formulation and Composition in Phenolic Resin Friction Materials

3.1 Typical Formulation Components

Phenolic resin friction materials containing zinc oxide whiskers typically comprise the following components:

Component	Function	Typical Content
Phenolic resin binder	Matrix, bonding	5–30 wt%
Zinc oxide whiskers	Reinforcement, functional filler	1–10 wt%
Reinforcing fibers	Mechanical strength	20–30 wt%
Abrasives	Friction modification	15–32 wt%
Lubricants	Wear reduction, friction stabilization	9–15 wt%
Other fillers	Cost reduction, property tuning	Balance

A representative formulation for a wind turbine yaw friction plate using tetra-needle zinc oxide whiskers comprises: 5–8 parts adhesive (phenolic resin or modified phenolic resin), 20–28 parts reinforcing fiber, 15–32 parts abrasive, 9–15 parts lubricant, and 25–40 parts regulator (including T-ZnOw, magnesium oxide, and friction powder) .

3.2 Modified Phenolic Resin Systems

For enhanced performance, modified phenolic resins are often preferred. These include:

- **Nitrile-butadiene rubber modified phenolic resin**: Improves flexibility and impact resistance
- **Silicon modified phenolic resin**: Enhances thermal stability
- **Boron-silicon synergistic modified phenolic resin**: Provides improved heat resistance and mechanical properties

3.3 Surface Treatment of Zinc Oxide Whiskers

To optimize dispersion and interfacial bonding, T-ZnOw may be surface-treated. Silane coupling agents (amino-silane, epoxy-silane) are commonly applied to improve compatibility with the phenolic matrix . More advanced approaches include SiO_2 coating on the whisker surface, which enhances dispersion and interfacial interaction with the resin .

4. Manufacturing Processes

4.1 Mixing and Compounding

The manufacturing process begins with thorough mixing of the phenolic resin with reinforcements, fillers, and T-ZnOw. High-speed mixers or Henschel mixers are employed to achieve uniform dispersion . For rubber-modified systems, internal mixing at 50–80°C for 10–30 minutes may be used .

A typical mixing sequence involves:

1. Premixing the resin with reinforcing fibers to ensure uniform dispersion

2. Adding fillers and functional additives

3. Finally adding the T-ZnOw under conditions that minimize shear damage to the whisker structure

4.2 Molding and Curing

The compounded mixture is then subjected to hot pressing. Typical conditions include:

- **Temperature**: 160–180°C

- **Pressure**: 10–30 MPa

- **Time**: Dependent on part thickness (typically 60–80 seconds per mm)

For compression molding of synchronizer rings, temperatures of 180±300°C are employed . The curing process may involve multiple stages to ensure complete crosslinking and optimal mechanical properties .

4.3 Post-Curing and Finishing

After demolding, friction materials typically undergo staged post-curing to complete the crosslinking reaction and relieve internal stresses. A representative heat treatment schedule includes:

- 162°C for 3 hours
- 140°C for 1.6 hours
- 185°C for 3 hours

Final machining operations include surface grinding, grooving, and dimensional finishing to meet specifications .

5. Performance Characteristics

5.1 Friction and Wear Performance

The incorporation of T-ZnOw into phenolic resin friction materials has been demonstrated to improve both friction stability and wear resistance. In wind turbine yaw friction plates, formulations containing T-ZnOw achieved friction coefficients of 0.42–0.44 at 100°C, with wear rates of 0.10–0.11 × 10^{-7} $cm^3/N·m$. These values compare favorably with imported reference materials (friction coefficient 0.41, wear rate 0.16 × 10^{-7} $cm^3/N·m$) and original formulations (friction coefficient 0.43, wear rate 0.16 × 10^{-7} $cm^3/N·m$, with serious noise >90 dB) .

5.2 Noise Reduction

A particularly notable benefit of T-ZnOw in friction materials is the elimination of friction noise. The three-dimensional whisker structure disrupts the formation of vibration modes that generate squeal and chatter, resulting in quieter braking operation .

5.3 Thermal Properties

The thermal conductivity of T-ZnOw contributes to improved heat dissipation in friction materials. When combined with boron nitride flakes in phenolic formaldehyde composites, T-ZnOw creates a synergistic effect that dramatically enhances thermal conductivity. With 30 wt% BN and 30 wt% T-ZnOw, in-plane thermal conductivity reaches 1.96 W/m·K, which is 6.8 times higher than neat phenolic resin . This thermal management capability helps maintain stable friction performance under repeated or severe braking conditions.

5.4 Mechanical Properties

The addition of T-ZnOw improves the mechanical integrity of phenolic friction materials. Flexural strength increases significantly when T-ZnOw is used in combination with other fillers. The three-dimensional whisker network provides isotropic reinforcement that resists crack propagation and improves impact resistance .

6. Summary

Tetrapod-shaped zinc oxide whiskers offer a multifunctional approach to enhancing phenolic resin friction materials. Their unique three-dimensional geometry provides isotropic reinforcement, thermal management, and noise reduction capabilities that are difficult to achieve with conventional fibrous or particulate fillers. When properly dispersed and bonded within the phenolic matrix, T-ZnOw contributes to stable friction coefficients, reduced wear rates, and quieter operation. These characteristics make T-ZnOw-reinforced phenolic composites attractive for applications ranging from automotive brake pads to wind turbine yaw friction plates and electromagnetic brake components .